

MENTHOL

THE SCIENCE OF SENSATION

1 COMPOUND IDENTITY

Common name: Menthol

IUPAC name:

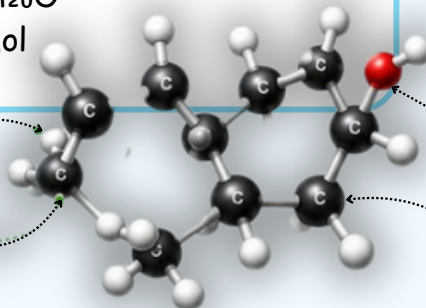
(1R,2S,5R)-2-isopropyl-5-methylcyclohexan-1-ol

Molecular formula: $C_{10}H_{20}O$

Molar mass: 156.27 g/mol

METHYL GROUP
(-CH₃)

ISOPROPYL GROUP
(-CH(CH₃)₂)



2 STRUCTURE AND IDENTITY



Common name: Menthol



Functional group: Alcohol (-OH)



Notable feature: Chiral molecule with 3 stereogenic centers

HYDROXYL GROUP
(-OH)

CYCLOHEXANE RING
(C₆H₁₁)

3 IMPORTANT PROPERTIES



Colorless/white crystalline solid



Strong, refreshing minty odor



Melting point: about 42-44 °C

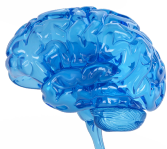


Slightly soluble in water



Moderately polar due to its hydroxyl (-OH) group

4 WHAT MAKES IT AMAZING?



Menthol activates TRPM8 receptors, which normally respond to cool temperatures. This sends a "cold" signal to the brain, creating a refreshing cooling sensation.

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RELEVANCE TO ENGINEERING

Menthol is important in product development and manufacturing. Engineers apply its chemical and physical properties when designing processes for extraction, purification, formulation, cooling products, and quality control.

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FUN FACT

Menthol is the compound that gives mint its coolness, but the actual temperature does not change!



3 USES & VALUE



Medicines: Used in creams, ointments, lozenges, and topical products for its cooling and soothing effects.



Oral care: Added to toothpaste and mouthwash for its refreshing mint flavor and cooling sensation.



Food & flavoring: Used to provide a strong mint flavor in chewing gum, candies, and other products.



Cosmetics & personal care: Found in lotions, balms, shampoos, and similar products for its cooling sensation and minty fragrance.



Fragrances: Valued for its distinctive minty aroma in perfumes and other scented products.

6 RISK AND SAFETY

Skin & eye irritation: Concentrated menthol may cause irritation or a burning sensation.

Ingestion hazard: Large amounts can be harmful if swallowed.

Respiratory irritation: High concentrations may irritate the nose and airways.

Keep away from children: Concentrated menthol products should be kept out of reach of children.

Safe handling: Avoid contact with eyes and prolonged contact with concentrated menthol; wash hands after handling.

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REFERENCES

PubChem. (n.d.). Menthol. National Center for Biotechnology Information.

National Center for Biotechnology Information. (n.d.). Menthol compound summary.

